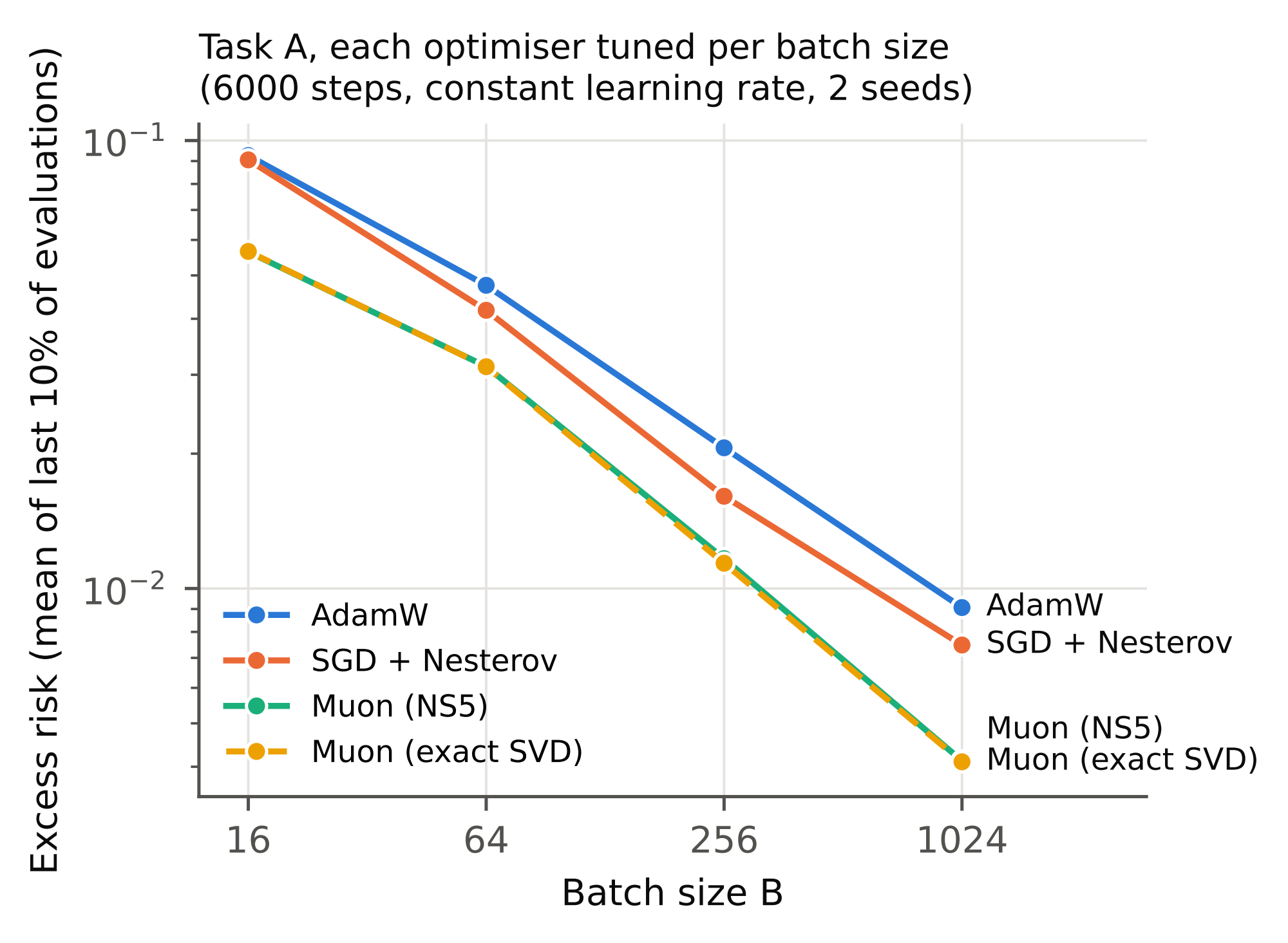
Task A, each optimiser tuned per batch size
(6000 steps, constant learning rate, 2 seeds)
Excess risk (mean of last 10% of evaluations)
10^{-1}
10^{-2}
16
64
256
1024
Batch size B
AdamW
SGD + Nesterov
Muon (NS5)
Muon (exact SVD)
AdamW
SGD + Nesterov
Muon (NS5)
Muon (exact SVD)